

Federico Nutarelli

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PERSONAL DETAILS

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ACADEMIC APPOINTMENTS

Long-term Visiting Scholar, University of London - City St. George	September 2026 - Currently
Long-term Visiting Scholar, Toulouse School of Economics	September 2025 - Currently
Visiting Scholar, MIT Sloan School of Management	March 2024 - September 2024
Assistant Professor in Economics, IMT School Lucca	September 2023 - Currently
Postdoctoral Fellow, Bocconi University	June 2022 - January 2024
Research Assistant, IMT School for Advanced Studies	May 2022
<i>Under the supervision of Prof. M. Riccaboni</i>	

EDUCATION

PhD in Economics, Networks and Business Analytics <i>IMT School for Advanced Studies</i>	2017-2022
Master of Science in Economics <i>Sant'Anna School for Advanced Studies, 110/110 Summa cum Laude</i>	2017
Bachelor Degree in Economics and Business <i>University of Pisa, 110/110 Summa cum Laude</i>	2015

ADDITIONAL EDUCATION

Course in Empirical IO II (Prof. Chuqing Jin) <i>Toulouse School of Economics (TSE)</i>	Spring 2025
Course in Empirical IO I (Prof. Dubois, Prof. Reynaert) <i>Toulouse School of Economics (TSE)</i>	Fall 2025
Course in User Innovation (Prof. Von Hippel) <i>MIT Sloan</i>	Spring 2024
Summer School <i>SIDE (Italian Econometrics Association), Bertinoro, "Machine Learning Algorithms for Econometricians"</i>	2019
Summer School <i>TU Wien, "Machine Learning Methods and Data Analytics in Risk and Insurance" (VISS2018).</i>	2018

AWARDS AND SCHOLARSHIPS

Principal Investigator (PI) <i>Innovation Diffusion: a consumer-side perspective.</i> <i>OpenLab Winner for Excellence research in Italy (MIUR) 2023</i>	2023-Currently
Scholarship Roche Holding AG <i>Role: research assistant</i>	2020-2023
Indam GNAMPA 2020 Project Funding <i>Role: co-investigator</i>	2020
Full Doctoral Scholarship, IMT School for Advanced Studies <i>Awarded with the highest score out of 3100 candidates (ex aequo)</i>	2017
ISTAT <i>Selected among the top 15 students</i>	2016
University of Pisa -Sant'Anna School for Advanced Studies <i>Selected as a student in MsC Economics</i>	2015

AWARDED PROJECTS

PRIN project: "Policy Impact for Pharmaceutical Innovation (PI²)" Currently
CASD project: "Machine Learning Methods for Studying Firms' Responses to Unfavorable Trade Shocks"
2023-Currently

RESEARCH INTERESTS

Empirical Industrial Organization, Pharmaceutical Industry, Causal Machine Learning

WORKS IN THE MAIN FIELD

Published

- * *Optimal data collection design in machine learning: the case of the fixed effects generalized least squares panel data model.* (with G. Gnecco and Selvi D.), **Machine Learning**, 2021, DOI: 10.1007/s10994-021-05976-x
- * *"Assessing the Heterogeneous Impact of Economy-Wide Shocks: A Causal Machine Learning Approach Applied to COVID-19"* (with Dueñas M., Riccaboni M., Ortiz V., Serti F.). **Oxford Bulletin of Economics and Statistics**: 1–28, <https://doi.org/10.1111/obes.70133>.

Complete Drafts

- * *"Demand Responsiveness and International Pharmaceutical Pricing"*
- * *"Product recalls, market size and innovation in the pharmaceutical industry"* (with Riccaboni M.) under revision in **Journal of the European Economics Association (JEEA)**
- * *"Learning to Import through Production Networks"* (with Huremovic K., Serti F., Vega-Redondo F.), submitted to the **Journal of International Economics**

Works in Progress

- *"Optimal time for recall? A study in the pharmaceutical market."*
- *"The microeconomic consequences of a unified European Scientific board in evaluating ATMPs"* (with Prof. Dubois)
- *"A Neyman orthogonal test of conduct"* (with Prof. Daniel Silva Junior)

WORKS IN OTHER FIELDS

* Indicates the publications where I contributed as first author or the co-first author.

Journal Publications (Peer Reviewed)

- 2016 * *"Testing the quality of the delineation of Local Labour Market Areas"* (with Reale A. and Drago C.) - internal publication Italian Institute of Statistics (ISTAT)
- 2020 * *Optimal trade-off between sample size, the precision of supervision, and selection probabilities for the unbalanced fixed effects panel data model.* (with G. Gnecco and Selvi D.), **Soft Computing**, 2020, 24, 15937–15949
- 2019 * *On the trade-off between number of examples and precision of supervision in machine learning problems* (with G. Gnecco), **Optimization Letters**, DOI: 10.1007/s11590-019-01486-x
- 2022 * *A machine learning approach to economic complexity based on matrix completion* (with M. Riccaboni and G. Gnecco), **Nature-Scientific Reports**, DOI: <https://doi.org/10.1038/s41598-022-13206-0>
- 2023 * *"The matrix completion of world trade: an analysis of interpretability through Shapley values"* (with Gnecco G., Riccaboni M.), **The World Economy**, DOI: 10.1111/twec.13457
- 2025 * *"Predicting the technological complexity of global cities based on unsupervised and supervised machine learning methods"*, **Journal of Economic Behavior & Organization**, DOI: 10.1016/j.jebo.2025.107011
- 2025 * *"Modeling Innovation Ecosystem Dynamics through Interacting Reinforced Bernoulli Processes"* (with Aletti G., Crimaldi I., Ghiglietti A.). **Research Policy**, DOI: <https://doi.org/10.1016/j.respol.2026.105587>.

TEACHING

Lecturer at IMT School for Advanced Studies <i>Graduate course: Labor Economics</i>	October 2026
Lecturer at IMT School for Advanced Studies <i>Graduate course: Introduction to R and STATA</i>	Fall 2026 (TBD)
Lecturer at IMT School for Advanced Studies <i>Graduate course: Labor Economics</i>	October 2025
Lecturer at IMT School for Advanced Studies <i>Graduate course: Machine-Learning and Causal Inference</i>	June 2023
Tutoring at Sant'Anna School for Advanced Studies <i>Undergraduate course: Advanced Econometrics</i>	March 2022-May 2022
Lecturer at IMT School for Advanced Studies <i>Graduate course: Applied Data Science</i>	2020-2021
Lecturer at IMT School for Advanced Studies	2022-2023

CONFERENCES AND INVITED SEMINARS

2025	Internal Seminar Sewries (Dept. Economics), Universidad de Chile (FEN)
2025	CFE-CMStatistics 2025, Birkbeck University of London
2025	30th International Panel Data Conference (IPDC), Université de Montpellier
2023	MIT Fall Innovation Lab, MIT
2023	"Complex system approaches to 21st century challenges: inequality, climate change, and new technologies", Santa Fe Institute.
2023	DRUID 2023, Nova Business School, Portugal.
2023	EAI Intetain 2023 Held online. Role: Organizer
2022	Research Policy (RP) 4 th conference for Early Career Researchers Held online.
2022	DRUID 2022, Copenhagen Business School, Denmark.
2022	ASSA 2022, American Economic Association.
2021	YETI 2021, YETI, Young Economists of Tuscan Institutions Conference, Pisa
2021	AMASES 2021, "Networks, Big Data, and Artificial Intelligence in Economics, Finance, and Social Sciences" Parallel Session.
2021	ODS 2021 "International Conference on Optimization and Decision Science (ODS2021)" Parallel Session
2020	ISOR Kolloquium, Department of Statistics and Operations Research of the University of Wien, Organizer: Immanuel Bomze.
2019-2020	IMT Reading group, Presented several machine learning papers as a speaker at the "Reading Group in Machine Learning and Networks".
2019	LOD 2019, speaker as The Fifth International Conference on Machine Learning, Optimization, and Data Science (LOD2019)
2019	ODS 2019, in Program of the International Conference on Optimization and Decision Science (ODS 2019), p. 16, Genoa, Italy, September 4 th -7 th, 2019.
2019	AMASES 2019, in Program of the 43rd Annual Meeting of the Italian Association for Mathematics Applied to Social and Economic Sciences (AMASES 2019), p. 4, Perugia, Italy, September 9 th -11 th, 2019.

EMPLOYMENT OUTSIDE ACADEMIA

Data Analysis <i>Roche</i>	2020-2023
Supervision : Under Prof. Dr. Riccaboni M. and Giuliani G. About : Analysis of innovation in the biotechnological context and in the enhancement of advanced therapies.	
Data Science <i>LVMH, Milan</i>	2019
Supervision : Under Dr. Romboli F. About : Efficient storage of the company data and click through rate analysis.	
Networks <i>ISTAT (Italian National Institute of Statistics)</i>	2016
Supervision : Under Dr. Reale A. and Prof. Dr. Drago C. About : Analysis of complex systems and Network theory.	

LANGUAGES

Italian	Mother tongue
English	Full Professional Proficiency
German	Advanced

COMPUTER SKILLS

<i>Programming Languages</i>	PYTHON
<i>Statistical Software:</i>	R, STATA, MATLAB
<i>Mark up languages</i>	L ^A T _E X, Beamer
<i>Other Software</i>	GIT/GITHUB

EXTRA CURRICULAR ACTIVITIES

1. Member of Chess Club
2. Boxe
3. Guitar